DATA & ANALYTICS ENGINEER

TECHNICAL ASSESSMENT
CANDIDATE TAKE-HOME ASSIGNMENT

OVERVIEW

This assessment evaluates your ability to design and deliver an end-to-end analytics solution similar to those our team builds internally. You will construct a dimensional data mart from an operational source, load it through SSIS with a full Slowly Changing Dimension implementation, and deliver a Power BI dashboard that surfaces key operational KPIs to terminal management.

The scenario is representative of real work performed by our Data & Analytics team. We are less interested in decoration and more interested in how you think about grain, change management, data quality, and the relationship between modelling decisions and the analytical questions they must answer. Clearly documented reasoning is valued as highly as a working build.

ITEM	DETAIL
Role	Data & Analytics Engineer
Deliverables	SSIS data mart (Part 1) and Power BI dashboard (Part 2)
Provided Files	PortOps_SourceData.xlsx containing 9 sheets
Environment	SQL Server 2019+ (Developer or Express), SSDT / Visual Studio, Power BI Desktop
Deadline	5 business days from receipt
Estimated Effort	10 – 14 hours of focused work
Submission Format	Single .zip file with README, SSIS solution, SQL scripts, .pbix, and screenshots

BUSINESS SCENARIO

You are supporting a container terminal operator. Operations leadership needs reliable daily visibility on container throughput, vessel turnaround, gate activity, equipment utilisation, and customer-level performance. Today this information lives across multiple operational systems and is consolidated manually into spreadsheets, which creates delays, version drift, and inconsistent figures across teams.

A consolidated export of the operational data has been provided as a single Excel workbook (PortOps_SourceData.xlsx). Read the README sheet inside the workbook first — it describes each source sheet and flags known quirks. Your task is to translate this raw data into a governed data mart and a decision-ready dashboard that management can rely on as a single source of truth.

SOURCE WORKBOOK STRUCTURE

SHEET	ROW COUNT (APPROX.)	PURPOSE
VesselCalls	~1,000	One row per vessel call (arrival, departure, planned vs. actual moves)
ContainerMovements	~90,000	Main operational fact — one row per container move
GateTransactions	~97,000	One row per truck gate transaction; contains gate-in and gate-out timestamps
Customers	12	Current state of customer master
CustomerHistory	Variable	Historical tier and credit-limit changes — the source for SCD Type 2
Terminals	4	Terminal and yard locations
Equipment	12	Cranes, RTGs, reach stackers, terminal tractors
Shifts	3	Operational shift patterns
README	N/A	Data dictionary and notes

PART 1 — DATA MART (SSIS) | 60%

OBJECTIVE

Design and implement a dimensional data mart in SQL Server and load it from the provided Excel workbook using SSIS. The mart must serve three analytical areas: operational throughput, vessel and berth performance, and gate activity. Your design should balance query performance, maintainability, and the ability to track historical change over time.

DATA MART REQUIREMENTS

At minimum, your mart must contain the following structures. You may add dimensions or facts if they strengthen the model — document your reasoning.

OBJECT	TYPE	NOTES
dim_date	Dimension	Full date dimension covering the reporting period. Include fiscal attributes; the company fiscal year begins 1 April.
dim_customer	Dimension	SCD implementation required. Source the change history from the CustomerHistory sheet.
dim_terminal	Dimension	Type 1 — overwrite on change.
dim_equipment	Dimension	Type 1. Include equipment type and capacity.
dim_shift	Dimension	Type 1. Small static dimension.
fact_container_movement	Fact	Grain: one row per container move. Include a derived measure for crane cycle time in seconds.
fact_vessel_call	Fact	Grain: one row per vessel call. Include berth delay, stay hours, and planned-vs-actual moves variance.
fact_gate_transaction	Fact	Grain: one row per gate transaction. Must carry two separate date foreign keys — gate-in and gate-out — to enable the Power BI inactive-relationship work in Part 2.

SLOWLY CHANGING DIMENSION

dim_customer is the focal point of your SCD work. The Customers sheet shows only current state; the CustomerHistory sheet contains the change log. Treat the attributes as follows:

- customer_tier and credit_limit — track history. A Type 2 implementation is expected.
- customer_name, customer_code, country, active_flag — overwrite on change (Type 1).

Your SCD Type 2 implementation must not rely on the built-in SSIS SCD Wizard. Build the pattern explicitly using Lookup, Conditional Split, and destination components. Be ready to justify this choice, including the performance characteristics of the Wizard versus a manual implementation. A working Type 1 implementation meets the minimum bar but will limit the ceiling on your Part 1 score.

ENGINEERING EXPECTATIONS

- Separate staging and target layers. Data lands in staging tables first; facts load only after all required dimensions have completed.
- Parameterise external paths (source workbook location, connection strings) so the solution is portable across environments.
- Implement surrogate keys on all dimensions. Fact tables must reference dimensions via surrogate keys, not natural keys — this is especially important for dim_customer where SCD Type 2 creates multiple rows per natural key.
- Handle unknown or late-arriving members explicitly. A default row (surrogate key = -1) in each dimension is one accepted pattern.

- Configure error-handling paths on Lookup and Data Conversion components so failed rows are captured to an error table rather than silently dropped.
- Log every package execution to an audit table with start time, end time, row counts per table, and status.

DATA QUALITY CHECKS

Embed data-quality validation in the pipeline. At a minimum, implement:

- Row-count reconciliation between source, staging, and target. Mismatches must halt or flag the load, not be ignored.
- Referential integrity — every fact row must resolve to a valid dimension member, including your unknown row.
- At least one business-rule check you consider relevant (for example, a container move cannot end before it starts, or a delivered date cannot precede a pickup date).
- Null, whitespace, and data-type validation where the source workbook is known to be inconsistent.

DOCUMENTATION (DESIGN_DOC.MD)

A short design document is expected alongside your SSIS solution. Address the following:

- The grain of each fact table and why you chose it.
- Your SCD Type 2 pattern — which components detect change, how historical rows are expired, how the current flag is maintained.
- How dim_date was populated and the range chosen.
- Data quality issues you encountered in the source and how you handled each.
- One or two changes you would make if the source grew from ~90K rows to 10M+ rows.

PART 2 — POWER BI DASHBOARD | 40%

OBJECTIVE

Build a Power BI report on top of the data mart you delivered in Part 1. The audience is operations management, who consume it daily to check performance, investigate exceptions, and hold stakeholders accountable against targets. Clarity, trend direction, and exception highlighting take priority over visual embellishment.

MODEL REQUIREMENTS

- Connect to the data mart in Import mode. Preserve the star schema in the Power BI model — dimensions filtering facts, single-direction relationships unless bi-directional is explicitly justified in your notes.
- Mark dim_date as the Date Table in Modeling settings. All time-intelligence calculations must flow through it.
- Relate fact_gate_transaction to dim_date twice — one active relationship on gate-in date, one inactive relationship on gate-out date. Do not work around this by importing dim_date twice.
- Use surrogate keys, not natural keys, for the fact-to-dimension relationships.

REQUIRED DASHBOARD CONTENT

Three report pages, each with a clear purpose:

PAGE 1 — OPERATIONS OVERVIEW

- KPI cards covering container moves, average crane cycle time, berth occupancy, and gate transaction volume.

- A trend visual showing daily container moves over the reporting period, with a rolling average to smooth daily variance.
- A breakdown of moves by terminal and by shift.
- Date and customer slicers that propagate across all pages.

PAGE 2 — GATE PERFORMANCE

- Comparison visual showing daily gate-ins and gate-outs on the same time axis. When the user changes the date slicer, the gate-outs line must respond according to gate-out date, not gate-in date.
- Average truck turnaround time with appropriate distribution (for example, buckets of 0–30, 30–60, 60–90, 90+ minutes).
- A table or matrix listing customers by gate volume with their average turnaround time.

PAGE 3 — CUSTOMER & VESSEL PERFORMANCE

- Top 10 customers by container volume with year-on-year comparison.
- Vessel call list showing berth delay, stay hours, and planned-vs-actual moves variance, sorted to surface underperformers.
- A visual that demonstrates your SCD implementation is working correctly. For example, customer volume split across the historical tiers they held during the reporting period. If you delivered only SCD Type 1, state that plainly and show current tier instead.

REQUIRED DAX MEASURES

Document every measure in dax_measures.md with the DAX code and a one- to two-sentence explanation of its logic and filter context behaviour.

MEASURE	NOTES
Total Container Moves	Core fact count.
Avg Crane Cycle Seconds	Must handle divide-by-zero correctly using DIVIDE.
Gate-Ins Count	Uses the active relationship on gate-in date.
Gate-Outs Count	Uses the inactive relationship on gate-out date, activated through USERELATIONSHIP.
Avg Truck Turnaround Minutes	Must be filtered by gate-out date, not gate-in date.
Moves YoY %	Year-on-year growth using a proper time-intelligence pattern.
Moves 7-Day Rolling Avg	Calculated with DATESINPERIOD or an equivalent pattern.
Berth Delay Avg Hours	Operates at the vessel-call fact grain.

QUALITY EXPECTATIONS

- Measures, not calculated columns, for aggregation. Calculated columns only where they materially simplify the model (for example, a grouping flag used in a slicer).
- DIVIDE for all division. VAR for readability in non-trivial measures.
- Meaningful measure names. No "Measure 1" or unnamed card values.
- Consistent visual formatting, sensible number formats, and clear titles.
- Performance — the report should respond to slicers within a few seconds on this dataset.

SUBMISSION

Package your work as a single .zip file named submission_<YourFullName>.zip structured as below. Please do not include database backups — scripts are sufficient.

FOLDER / FILE	CONTENT

README.md	Setup steps, tool versions used, assumptions, and answers to the written questions.
part1_datamart/ssis/	SSIS solution (.sln + .dtsx packages).
part1_datamart/sql/	DDL for staging and mart objects, plus date-dimension seed script if done in SQL.
part1_datamart/design_doc.md	Design rationale, SCD approach, data quality handling, and trade-offs.
part1_datamart/screenshots/	Control flow, data flow, SCD component, and sample target-table output.
part2_powerbi/dashboard.pbix	The Power BI report file.
part2_powerbi/dax_measures.md	All measures with code and explanations.
part2_powerbi/screenshots/	Each report page and a screenshot of the model view.

WRITTEN QUESTIONS

Answer the following in your README.md. Keep each answer to 3–5 sentences. These questions assess depth of understanding and are weighted as part of the overall score. Generic textbook answers score lower than answers grounded in your actual implementation choices.

DATA WAREHOUSING

Explain the practical difference between SCD Type 1 and Type 2. Using examples from this assessment, justify where you applied each.

Why should a fact table reference a dimension via surrogate key rather than natural key? Give at least two reasons specific to your dim_customer implementation.

Your dim_date is bounded. What happens if a fact arrives with a date outside that range, and how would you design the pipeline to handle it without failure?

SSIS

Why is the built-in SSIS SCD Wizard not suitable for a production Type 2 load at scale? Give specific technical reasons.

Describe how you would implement automated row-count reconciliation between source, staging, and target — including what should happen when counts disagree.

What is the role of a staging layer in a data warehouse load? What would go wrong if you loaded the Excel file directly into the final fact tables?

POWER BI

Why does Power BI allow only one active relationship between two tables? What ambiguity would multiple active relationships create?

Explain the difference between USERELATIONSHIP and CROSSFILTER. When is each the right choice?

A business user reports that a KPI on the dashboard does not respect the date slicer. List the most likely causes in the order you would investigate them.

EVALUATION CRITERIA

Your submission is scored across the dimensions below. The rubric is shared deliberately; candidates who understand how they are assessed produce sharper, more focused work.

DIMENSION	WEIGHT
Data mart design and modelling	15%

SSIS implementation quality	20%
SCD implementation (Type 2 preferred)	15%
Data quality and documentation	10%
Power BI model and USERELATIONSHIP usage	20%
DAX quality and correctness	10%
Dashboard design and usability	5%
Written questions	5%

We value clarity of reasoning as highly as the final deliverable. A clean, well-documented partial solution is preferred over a polished one that cannot be explained. Candidates who meet the bar will be invited to a technical interview, where we will ask you to walk through your design, defend one of your trade-offs, and respond to a small live change request.

GROUND RULES

You may use official documentation, reference material, and your own notes. AI assistants should not be used to generate full solutions; you will be asked to explain your work in the interview.

State your assumptions at the top of README.md. Clearly-stated assumptions are respected; unstated ones can count against you.

If you get stuck on a specific item, document your thinking, skip it, and move on. A partial answer with clear reasoning scores better than a copied or guessed complete one.

For a single consolidated clarification question, reply to the email that sent you this assessment. We respond within one business day.